

## LEIPZIG HALLE, Germany

WMO: 104690

Lat: 51.4347N

Lon: 12.2397E

Elev: 136

StdP: 99.71

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.1      | -8.0 | -13.9                         | 1.1 | -9.9 | -10.9 | 1.5 | -7.6 | 14.4                  | 7.0  | 13.0 | 6.4  | 4.0                   | 70   | 0.590 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.2                   | 31.2            | 19.6 | 29.1 | 19.0 | 27.1 | 18.4 | 21.0                | 27.7 | 20.1 | 26.6 | 19.4 | 25.3 | 4.0                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.1                            | 14.1 | 22.6 | 18.1 | 13.2 | 22.0 | 17.1 | 12.4 | 21.5 | 61.4          | 28.0 | 58.6 | 26.4 | 56.1 | 25.0 | 24.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.6          | 10.2 | 9.0 | -13.1                      | 34.9 | 4.3                | 1.8 | -16.2                                              | 36.1 | -18.7      | 37.2 | -21.1      | 38.2 | -24.2      | 39.5 |    |    |
| (5)               |      |     | -13.4                      | 22.9 | 4.2                | 1.0 | -16.5                                              | 23.7 | -18.9      | 24.2 | -21.3      | 24.8 | -24.4      | 25.5 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.3  | 0.7   | 1.2   | 5.1   | 10.3  | 14.0  | 17.8  | 20.1  | 19.2  | 15.6  | 10.7  | 5.9   | 2.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 7.85  | 4.96  | 4.62  | 4.50  | 3.99  | 3.94  | 3.60  | 3.41  | 3.20  | 3.25  | 3.74  | 3.81  | 4.31 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1141  | 290   | 246   | 159   | 44    | 9     | 1     | 0     | 0     | 1     | 37    | 131   | 223  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3109  | 548   | 479   | 412   | 241   | 144   | 52    | 21    | 27    | 95    | 238   | 372   | 481  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1260  | 1     | 0     | 6     | 54    | 132   | 233   | 314   | 284   | 168   | 58    | 8     | 1    | (10) |
| (11) |                                                                                  | CDD18.3     | 188   | 0     | 0     | 0     | 1     | 9     | 36    | 77    | 52    | 13    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 1699  | 0     | 0     | 0     | 9     | 99    | 331   | 704   | 443   | 109   | 3     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 541   | 0     | 0     | 0     | 2     | 22    | 94    | 254   | 147   | 21    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.5   | 5.4   | 4.9   | 4.9   | 4.3   | 4.2   | 3.9   | 4.0   | 3.9   | 4.1   | 4.3   | 4.8   | 5.4  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 557   | 35    | 29    | 37    | 34    | 57    | 58    | 76    | 61    | 51    | 36    | 45    | 39   | (15) |
| (16) |                                                                                  | PrecMax     | 780   | 62    | 54    | 95    | 85    | 130   | 91    | 134   | 149   | 137   | 95    | 104   | 78   | (16) |
| (17) |                                                                                  | PrecMin     | 362   | 4     | 6     | 6     | 3     | 5     | 14    | 15    | 22    | 13    | 7     | 0     | 15   | (17) |
| (18) |                                                                                  | PrecStd     | 91    | 15    | 13    | 24    | 19    | 33    | 21    | 31    | 30    | 27    | 20    | 24    | 18   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 12.8  | 13.2  | 19.2  | 24.2  | 29.2  | 32.0  | 34.9  | 33.9  | 29.2  | 23.8  | 17.0  | 13.0 | (19) |
| (20) |                                                                                  |             | MCWB  | 9.9   | 7.9   | 11.9  | 15.0  | 18.9  | 20.3  | 20.8  | 19.9  | 18.3  | 16.7  | 13.9  | 10.1 | (20) |
| (21) |                                                                                  | 2%          | DB    | 10.8  | 10.9  | 15.9  | 22.0  | 26.1  | 29.2  | 31.9  | 30.2  | 26.2  | 20.1  | 14.2  | 11.1 | (21) |
| (22) |                                                                                  |             | MCWB  | 8.6   | 7.9   | 10.7  | 14.0  | 17.4  | 19.1  | 19.5  | 20.1  | 17.3  | 15.3  | 11.8  | 8.9  | (22) |
| (23) |                                                                                  | 5%          | DB    | 8.9   | 9.1   | 13.2  | 20.0  | 23.8  | 27.1  | 29.8  | 27.9  | 24.0  | 18.0  | 13.0  | 9.9  | (23) |
| (24) |                                                                                  |             | MCWB  | 7.4   | 6.7   | 9.0   | 13.4  | 16.4  | 18.7  | 19.1  | 18.8  | 17.0  | 14.3  | 10.8  | 8.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | 7.2   | 7.2   | 11.2  | 17.9  | 21.2  | 25.0  | 27.1  | 25.2  | 21.8  | 16.2  | 11.2  | 8.2  | (25) |
| (26) |                                                                                  |             | MCWB  | 6.0   | 5.2   | 8.0   | 12.4  | 15.0  | 17.5  | 18.3  | 17.9  | 16.1  | 13.5  | 9.3   | 6.9  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 10.5  | 9.6   | 12.7  | 16.0  | 20.1  | 22.5  | 21.9  | 22.3  | 20.0  | 17.3  | 14.1  | 10.7 | (27) |
| (28) |                                                                                  |             | MCDWB | 12.3  | 10.9  | 17.1  | 22.3  | 27.0  | 28.1  | 29.9  | 29.6  | 26.4  | 21.7  | 16.3  | 12.6 | (28) |
| (29) |                                                                                  | 2%          | WB    | 8.8   | 8.1   | 11.1  | 14.9  | 18.5  | 20.4  | 20.9  | 20.8  | 18.5  | 15.8  | 12.4  | 9.4  | (29) |
| (30) |                                                                                  |             | MCDWB | 10.4  | 10.3  | 14.7  | 20.6  | 24.5  | 26.9  | 28.0  | 27.3  | 23.7  | 19.0  | 14.3  | 10.9 | (30) |
| (31) |                                                                                  | 5%          | WB    | 7.2   | 6.7   | 9.8   | 13.9  | 17.1  | 19.3  | 20.1  | 19.9  | 17.5  | 14.7  | 11.0  | 8.1  | (31) |
| (32) |                                                                                  |             | MCDWB | 8.4   | 8.7   | 12.7  | 19.2  | 22.4  | 25.4  | 26.6  | 26.0  | 22.3  | 17.5  | 12.8  | 9.6  | (32) |
| (33) |                                                                                  | 10%         | WB    | 5.9   | 5.3   | 8.4   | 12.7  | 15.7  | 18.2  | 19.4  | 19.0  | 16.5  | 13.7  | 9.8   | 7.0  | (33) |
| (34) |                                                                                  |             | MCDWB | 6.9   | 7.1   | 11.1  | 16.9  | 20.1  | 23.6  | 25.4  | 23.8  | 20.6  | 16.1  | 11.2  | 8.1  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 4.7   | 6.1   | 7.9   | 10.1  | 9.9   | 10.2  | 10.2  | 9.9   | 9.1   | 7.5   | 5.5   | 4.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 5.9   | 8.4   | 11.6  | 13.6  | 13.5  | 13.6  | 14.0  | 13.9  | 12.9  | 9.9   | 7.2   | 6.0  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.6   | 5.8   | 7.1   | 7.3   | 6.5   | 5.8   | 4.9   | 5.3   | 5.8   | 5.9   | 5.2   | 4.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 5.5   | 7.8   | 10.5  | 12.5  | 12.3  | 12.3  | 11.5  | 11.7  | 10.8  | 9.1   | 6.6   | 5.6  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.6   | 5.6   | 6.8   | 7.1   | 6.2   | 5.8   | 4.6   | 5.2   | 5.6   | 5.8   | 5.1   | 4.7  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.317 | 0.341 | 0.384 | 0.417 | 0.419 | 0.415 | 0.433 | 0.426 | 0.388 | 0.366 | 0.340 | 0.311 | (40) |      |
| (41) |                                                                                  | taud        | 2.362 | 2.329 | 2.223 | 2.184 | 2.215 | 2.255 | 2.212 | 2.251 | 2.338 | 2.373 | 2.385 | 2.364 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 692   | 774   | 801   | 818   | 835   | 841   | 817   | 802   | 792   | 734   | 655   | 633   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 63    | 85    | 113   | 132   | 134   | 130   | 133   | 122   | 100   | 81    | 62    | 54    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.73  | 1.43  | 2.44  | 3.99  | 4.88  | 5.23  | 4.93  | 4.26  | 3.07  | 1.78  | 0.85  | 0.56  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.10  | 0.23  | 0.31  | 0.51  | 0.58  | 0.43  | 0.63  | 0.38  | 0.39  | 0.20  | 0.12  | 0.06  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (48 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | +0.37 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page